# BE THROUGH AII

## 3-Minute Hackathon Demonstration — FINAL

*Sustainability • Governed Circular Resource Decision Intelligence*

**AI RECOMMENDS → HUMANS VERIFY → THE ENTERPRISE DECIDES**

## Before You Record

- Open the deployed BE THROUGH AII v5.3 prototype and start from the Decision Cockpit.
- Record the screen plus your voice. Move the cursor deliberately and pause briefly on key values.
- Keep “SIMULATED VERIFICATION STATE” visible on prototype verification data.
- Target 2:45–3:00. Show the decision chain, not every dashboard field.

## Exact 3-Minute Demonstration

### 0:00–0:20 | OPENING / HOME

**SAY:** “Organic waste becomes a decision problem when multiple pathways compete for the same material and safety information may be incomplete. BE THROUGH AII puts AI inside a governed decision architecture: AI recommends, humans verify, and the enterprise decides.”

**SHOW:** Project name + Sustainability category + governance principle.

**TAKEAWAY:** This is governed AI, not autonomous plant control.

### 0:20–0:45 | DECISION COCKPIT — BASELINE

**SAY:** “For this 24-hour demonstration, 500 tonnes enter the facility. The deterministic engine accounts for every tonne: 150 tonnes to Feed, 250 to Biogas, and 100 to Compost. The boundary balances exactly to 500 tonnes, with zero unexplained discrepancy.”

**SHOW:** 500.00 t intake; 150 Feed; 250 Biogas; 100 Compost; 0 Hold; 0 Unaccounted; $49,250 gross modeled yield.

**TAKEAWAY:** The core allocation is mathematically controlled.

### 0:45–1:05 | 11-STAGE PIPELINE

**SAY:** “The system separates feasibility, decision support, and authorization. Deterministic rules validate first. AI interprets only the validated state. Feasible does not mean optimal, and modeled does not mean authorized.”

**SHOW:** Stages 01–06 completed; Stage 07 Human Verification active; FEASIBLE ≠ OPTIMAL ≠ AUTHORIZED.

**TAKEAWAY:** AI cannot turn its own recommendation into execution.

### 1:05–1:35 | SCENARIO C — DIGESTER MAINTENANCE

**SAY:** “Now I introduce a physical disruption. Digester capacity falls to 140 tonnes per day. The engine recalculates without violating pathway limits: 180 Feed, 140 Biogas, 150 Compost, and 30 tonnes in Controlled Hold. The full 500 tonnes remain accounted for.”

**SHOW:** Scenario C: 180 / 140 / 150 / 30 Hold; mass = 500 t.

**TAKEAWAY:** The system adapts to physical constraints instead of forcing an unsafe answer.

### 1:35–1:55 | MARGINAL TRADE-OFF

**SAY:** “The AI layer can explain a trade-off without making the decision for the enterprise. A one-tonne shift from Compost to Feed changes modeled gross value by 95 dollars, subject to quality, capacity, commercial acceptance, and human authorization.”

**SHOW:** Marginal engine: 1 t Compost → Feed; +$95 modeled gross delta.

**TAKEAWAY:** AI provides decision intelligence, not command authority.

### 1:55–2:20 | STAGE 07 — HUMAN VERIFICATION

**SAY:** “Before execution, humans independently verify the real-world state: laboratory quality, intake custody, digester telemetry, compost readiness, and hold conditions. These values are prototype demonstration fixtures, not live plant certifications.”

**SHOW:** Five verification classes with SIMULATED VERIFICATION STATE labels.

**TAKEAWAY:** Human verification is explicit and independent of AI.

### 2:20–2:45 | FAILURE CASE — SCENARIO E OR F

**SAY:** “If required digester telemetry is lost, the system refuses to invent a value. The remaining safe pathways can absorb only 430 tonnes, so intake acceptance is stopped. With conflicting sensor values, the affected pathway is blocked and field verification is required.”

**SHOW:** Scenario E: telemetry loss → intake stopped; or Scenario F: sensor conflict → pathway blocked.

**TAKEAWAY:** Missing information creates a controlled restriction, not a hallucinated answer.

### 2:45–2:58 | HARDWARE INTERLOCK

**SAY:** “Even with a valid recommendation, physical actuation remains inhibited. Execution requires enterprise authorization, the physical key permissive, and local safety permissives. An AI attempt to bypass this path is structurally rejected.”

**SHOW:** ACTUATION INHIBITED; authorization missing; physical key; E-stop; fieldbus relay.

**TAKEAWAY:** AI cannot directly actuate the plant.

### 2:58–3:00 | CLOSING

**SAY:** “500 tonnes in. Every tonne accounted for. Invalid states rejected. Missing information exposed. Human authority preserved. AI recommends. Humans verify. The enterprise decides.”

**SHOW:** Final Covenant screen.

**TAKEAWAY:** Finish on governance, not on a financial claim.

## Recording Checklist

- Show the working prototype, not only the slide deck.
- Show one normal case and at least two stress/failure cases.
- Explicitly show the human-governance and hardware-interlock boundary.
- State that sensor, operator, certificate, and timestamp values are synthetic demonstration fixtures.
- Keep the video at or under 3 minutes.
- Upload it to the public YouTube, Vimeo, or Google Drive location required by the hackathon.

## Submission-Safe Facts

- Canonical mass balance: 500.00 t = 150.0 Feed + 250.0 Biogas + 100.0 Compost + 0.0 Hold + 0.00 Unaccounted.
- Declared pathway ceilings: Feed 180 t/day; Biogas 300 t/day; Compost 150 t/day.
- Baseline gross modeled value: $49,250; modeled, not net profit.
- Baseline carbon displacement: 253.5 tCO2e; methodology-dependent modeled estimate.
- Maintenance: 180 Feed / 140 Biogas / 150 Compost / 30 Hold.
- Telemetry loss: remaining safe capacity 430 t, so intake acceptance is stopped rather than forcing an unsafe 500 t allocation.